Data Exploration and System Management - Key Concepts

Why do people collect data?

People collect data for three main reasons:

1. **Description of the surrounding reality** - This includes making an inventory of species, completing the description of processes, and estimating the impact of factors on the observed object.
2. **Hypothesis testing** - Testing whether something is true or false, for example "Does Santa Claus exist?"
3. **Prediction** - Forecasting future outcomes, such as predicting what the energy yield of a photovoltaic system installed at a selected site will be.

How do people collect data?

People collect data through three main methods:

1. **Simple, single observation** - Looking at what's going on without any structured approach.
2. **Representative inspection** - Selection of what will be observed, including standardisation of the object of observation.
3. **Experiments** - Manipulating the circumstances of the object under analysis and then observing the results, including standardisation of observation conditions.

What are the stages of data analysis?

The stages of data analysis are:

1. **Decision on what I want to know** - Defining the objective of the analysis.
2. **Data formatting** - Preparing data in the appropriate structure.
3. **Initial data exploration (data preprocessing)** - Exploring and cleaning the data before analysis.
4. **Essential data analysis (modelling)** - The core analytical work.
5. **Results presentation** - Communicating the findings.